

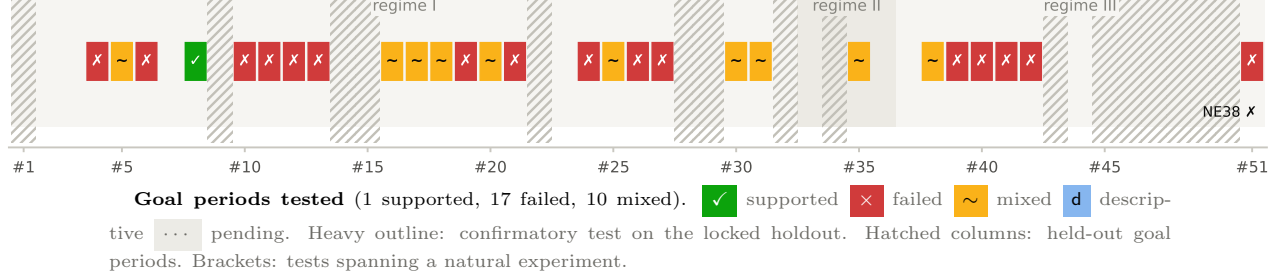
H125: The kickoff response is a damped oscillator: look for the day-2 undershoot

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 1.5/5

Bottom line. The kickoff response is overdamped: content overshoots the goal on day 1 and settles within two active days without swinging below it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH366 says the kickoff response is an underdamped oscillator: content overshoots the settled goal level on day 1 and swings below it on days 2–3. A fading kickoff field gives the overshoot without the swing. I measured each agent’s excess alignment with the kickoff on 27 kickoffs (regimes I–III, 4–21 agents) and compared days 2–3 with days 4–5, against 80 ordinary-day origins. The synthetic check came first: the test has size 0 and power 1 at damping $\zeta \leq 0.5$. Day 1 overshoots ($E_1 = +0.055 \pm 0.012$, 22/27). Days 2–3 do not undershoot: $U = -0.006 [-0.020, +0.008]$ (bge), $-0.002 [-0.017, +0.013]$ (gte), inside the placebo spread. The response is overdamped. Holdout: not run.

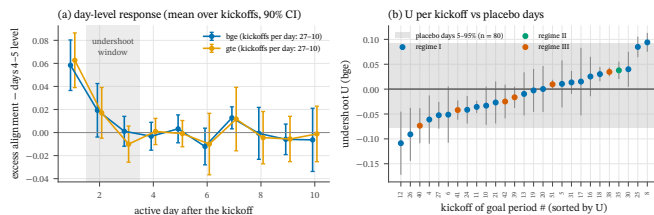
PHYSICAL MODEL

Model 11 with the overdamped relaxation of model 02 as the rival. Spin: agent i ’s statement vectors in the regime-whitened 32-d basis. Field: the kickoff, direction $\hat{k}$ from shared `goal_fields`. Order parameter: the excess alignment $a = \langle z, \hat{k} \rangle$ minus the mean over decoy kickoffs, averaged linearly (no renormalized windows). Clock: active hours h since the kickoff (H103). M_{osc} : damped oscillator; M_{fade} : overdamped spin in a fading field (two exponentials). Both carry a time-of-day profile.

$$M_{osc} : A(h) = A_{\infty} + Be^{-\lambda h} \cos(\omega h + \phi), \quad \zeta = \lambda / \sqrt{\lambda^2 + \omega^2}$$

$$M_{fade} : A(h) = A_{\infty} + c_1 e^{-h/\tau_1} + c_2 e^{-h/\tau_2}$$

$$U = \langle A_{i,4-5} - A_{i,2-3} \rangle_i, \quad \zeta_{pk} = \delta / \sqrt{\pi^2 + \delta^2}, \quad \delta = \ln(E_1/U)$$



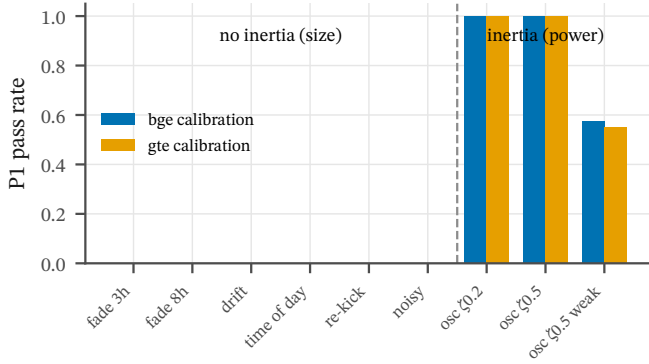
Observables. (a) Mean excess alignment relative to the days 4–5 level, by active day, over 27 kickoffs (both embedding models, 90% CI). Day 1 overshoots; day 2 keeps $+0.02$; day 3 and later sit at the settled level. The gray band marks where M_{osc} predicts an undershoot. (b) U per kickoff (bge) against ordinary-day origins (gray: 5–95%). Kickoffs do not exceed the placebo spread.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Read the steady state two active days after a kickoff. Day-1 content is about 0.055 above the settled alignment and day 2 about 0.02 above; there is no later dip to wait out. A drop in goal alignment on day 2 or 3 is the kickoff’s salience fading, not a loss of the goal. An operator who wants to know where a swarm settled can stop watching after day 3.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 undershoot $U > 0$, beyond placebo	bge $-0.006 [-0.020, +0.008]$, p 0.79; gte -0.002 , p 0.69	failed (kill)
P2 M_{osc} wins $\geq 2/3$	18/27, 19/27; null size 0.12–0.20	no weight
P3 $\zeta_{\text{pk}} < 1$	$\zeta_{\text{pk}} \geq 1$ in both models	failed
S1 overshoot $E_1 > 0$	$+0.055 \pm 0.012$ (22/27); gte $+0.062$	supported
S2 $U(\text{III}) > U(\text{I})$	-0.019 vs -0.010 , p 0.65	not supported
N1 G51 own roles ($N = 21$)	$U_{\text{own}} -0.009 / -0.026$	failed
N2 NE38 one agent	no overshoot, U at others' median	failed
N3 G38 ring-down	M_{osc} wins; size 0.19–0.26; ζ 0.56 vs 0.04	weak



Second figure. Synthetic swarms on the real skeleton of 27 kick-offs (40 replicates per world, both calibrations). P1 passes in no world without inertia (fading field, drift, time-of-day profile, day-4 re-kick, doubled day noise) and in every replicate with $\zeta = 0.2$ or 0.5 at the observed overshoot.

GOAL PERIODS IN DETAIL

G04 $\times$ failed
G05 $\sim$ mixed
G06 $\times$ failed
G08 $\checkmark$ supported
G10 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\sim$ mixed
G17 $\sim$ mixed
+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target is a text embedding of the kickoff, and day 1 contains restatements of it, which raise E_1 but not U . Only 10 long periods supply placebo origins. Damping between 0.5 and 1 gives an undershoot below $0.2 \times E_1$ and is not excluded. Model comparison by SSE is liberal for single kickoffs, so G38's preference for M_{osc} is weak. Two calendar days include village-off gaps that lengthen the active-hour clock.

ROUND 2

Fit the decay of the read-out spike (H127) on the call and hour clocks; remove kickoff restatements before measuring E_1 ; run `confirm.py` on the held-out kickoffs.

- **H125-R1** The overshoot at call resolution.
- **H125-R2** Restatement vs re-orientation.
- **H125-R3** G38's non-monotone 10 days.

H125-kickoff-damped-oscillator · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H125-kickoff-damped-oscillator/README.md · ratings: H125 agent (suggested) (page 2)